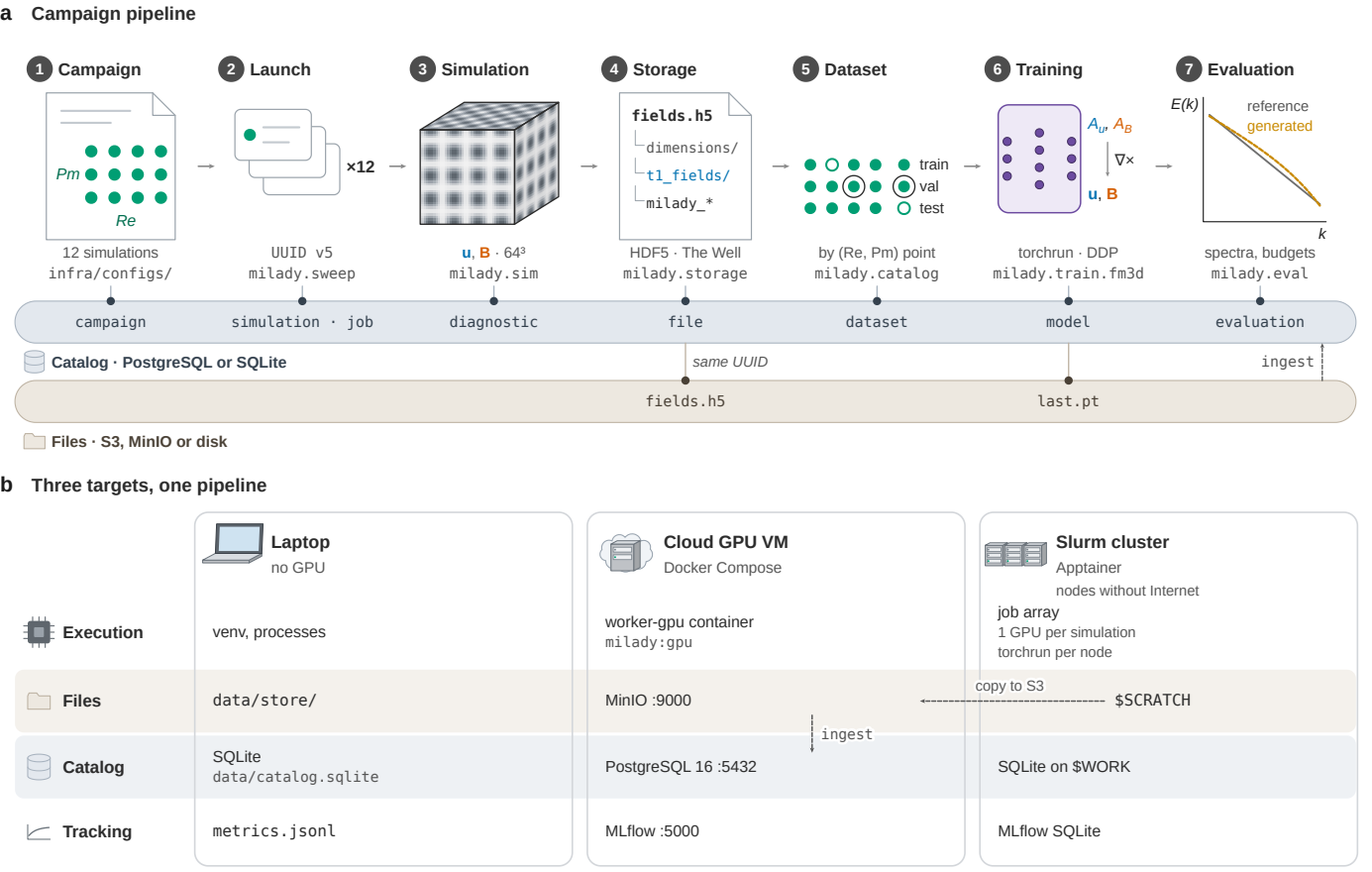

a Campaign pipeline
1 Campaign
2 Launch
3 Simulation
4 Storage
5 Dataset
6 Training
7 Evaluation
Pm
Re
×12
fields.h5
dimensions/
t1_fields/
milady_*
train
val
test
A_u, A_B
∇×
u, B
E(k)
reference
generated
k
12 simulations
infra/configs/
UUID v5
milady.sweep
u, B · 64³
milady.sim
HDF5 · The Well
milady.storage
by (Re, Pm) point
milady.catalog
torchrun · DDP
milady.train.fm3d
spectra, budgets
milady.eval
campaign
simulation · job
diagnostic
file
dataset
model
evaluation
Catalog · PostgreSQL or SQLite
same UUID
ingest
fields.h5
last.pt
Files · S3, MinIO or disk
b Three targets, one pipeline
Laptop
no GPU
Cloud GPU VM
Docker Compose
Slurm cluster
Apptainer
nodes without Internet
Execution
venv, processes
worker-gpu container
milady:gpu
job array
1 GPU per simulation
torchrun per node
Files
data/store/
MinIO :9000
copy to S3
$SCRATCH
ingest
Catalog
SQLite
data/catalog.sqlite
PostgreSQL 16 :5432
SQLite on $WORK
Tracking
metrics.jsonl
MLflow :5000
MLflow SQLite